Assessing the Viability of Haptic-Based Music Devices for the Deaf Community and Exploring Policy-Based Community Restrictions Leading to Music Inaccessibility

Introduction

Orthodox music is often characterized by lyricism, melodic behaviour, and harmonic resonance. Music focused on elements of rhythmic syncopation and harmonic dissonance is young, with genres of ragtime and early jazz pioneering such archetypes only in the late nineteenth and early twentieth centuries. Historically, music has often been inaccessible to deaf and hard-of-hearing communities. To these groups, music is often a multi-sensory engagement—ranging from standing near large speakers to feeling floor vibrations to observing visual interpretations. Despite advancements in technology and available resources, the integration of said technologies in such a field is unheard of. The proposal of a head-worn, wireless audio device which translates auditory frequencies and rhythmic patterns into vibrations offers a potential compromise to this gap in accessibility. Through interviewing deaf and hard of hearing individuals, this article aims to examine the viability of such a device through analyzing and weighing critical information provided through the lived experiences and preferences of these individuals.

The Argument for Viability: Tactile Connection and Privacy

The primary argument for a vibration-based device is the inherent way many DHH individuals already connect with music. Rosanna Ascenzo explains that the community enjoys music "through that way, in a different way than hearing people do," noting that "our body is there. We can feel a connection through our bodies to the music". Similarly, Sarahanne Dambrosio notes that while she cannot always make out words, she can "feel... the different beats".

DHH and music:

To determine the potential viability of such a proposal, it is imperative to understand the mechanisms which work best for the DHH. The most common form of complete deafness is sensorineural deafness, which is often due to irreversible hair cell damage in the cochlea, resulting in improper conversions of vibrations into nerve signals. Thus, the brain receives little to no auditory information. Regardless of a healthy connection between the cochlea and the auditory cortex, if hair cells are unable to work properly, then sound is unable to be interpreted in the traditional sense.

Drawing parallels to discover solutions:

To determine the solution which has the greatest potential, it is important to break down sound into its most foundational concept - waves. Soundwaves can be visualized and are often described as a vibration; Specific vibrations produce specific sounds. Using physical vibrations is a solution which utilizes a user’s working mechanoreceptors in their skin. Linear resonant

actuators can be programmed to vibrate a certain number of times per sample, or every twelve milliseconds, in order to produce frequencies, otherwise known as pitches, and their intensity is mapped directly to the loudness of different notes. Ultimately, the use of vibrations is a solution which allows for the replication of soundwaves, an attempt to maximise the authenticity of a musical experience.

Gauging interest and personal opinion in the proposal:

Addressing issues with current solutions:

Through conducting interviews, discoveries were made on the methods in which deaf and hard of hearing individuals use to enjoy musical experiences. With these discoveries, it was evident that current means of musical enjoyment conducted by the community have clear limitations that must be addressed.

Deaf and hard of hearing individuals often utilize their personal devices to access musical experiences. Through increasing volume levels to a maximum and contacting skin with device speakers, these individuals are able to feel the vibrations produced as a side product produced by sound waves. These vibrations are not only not specialized and not targeted, but are also the side product of loud sound. It is not uncommon that individuals use these methods in public spaces such as on public transportation or in crowded areas. Rosanna Ascenzo, a deaf individual shared her experience with using this method on a bus at maximum volume: “People were looking at me … it’s too loud … we don’t want to interrupt other people’s, you know, day to day”. A more localized vibration device which algorithmically translates soundwaves and sound files into vibrations would enable these individuals to experience a more private experience without needing to disturb others.

Concerts are also a way that deaf and hard of hearing individuals enjoy music. Concerts and similar venues utilize large speakers and subwoofers to blast music into a large, open space. These bass heavy vibrations are so strong that they can then be felt through the ground. However, by the time that these vibrations reach audience members, they are often dimmed down and muted, diminishing the effectiveness and meaningfulness of the experience. Sarahanne Dambrosio, another deaf individual, suggests that an ideal device would include “strong vibrations” and a “larger range when it comes to sort of interpreting different notes and different rhythms and loudness and quiteness”. Through this, Dambrosio suggests a desire for a more nuanced haptic language beyond the typical bass-heavy “thump” of a subwoofer. A tactile vibration device is able to provide a larger range of vibrations along with more targeted ones, increasing haptic variability and overall user experience.

Methods of music experience that deaf and hard of hearing individuals utilize are often stationary and not realistic for day-to-day usage. Hauling around a subwoofer is not feasible, not only for its size and weight, but also due to its disruptive nature. Blasting music on a personal

device is a public nuisance. A lightweight, bluetooth haptic vibration music device would be able to address all of these problems, providing a convenient, clean method of enjoying music.

Limitations of Vibration-Based Solutions:

Despite the clearly showcased interest in vibration-based music technology, several interviewees also expressed reservations about a solely vibration-based approach. Although convenient and enjoyable, vibrations are often insufficient at providing a full music experience, and can become overwhelming to a point of physical discomfort.

Vibration-based experiences are not always pleasant and when the fine line between enjoyment and overwhelming vibration is crossed, the experience can become physically uncomfortable, and sometimes even result in pain. An example of one such experience was described as not comfortable and even “headache” inducing. The device, a vibration vest, “picked up on the bass,” but the individual “actually didn’t like that.” as it “wasn’t comfortable”. This suggests that the placement and intensity of a head-worn vibration device would be a critical and sensitive design challenge.

For many individuals, the physical element of music—sound or vibration—is less important than understanding its content. One interviewee argues that “vibration is just a feeling… but with the meaning, with interpretation, and with the visuals, we can understand what the music is about”. Many other interviewees also shared similar views, prioritizing captions and sign language as the most important components of a musical experience.

These anecdotes pose a challenge to vibration-based devices. Not only can intense vibrations create physical discomfort, but they are also unable to provide literary context, which is highly valued in the deaf and hard-of-hearing community.

Design Considerations for Success:

For a head-worn vibration device to be viable, it must move beyond simple rhythmic pulses. The interviews suggest several "optimal" features. The device must have nuanced frequencies and rhythmic patterns, to create clear distinctions between notes and rhythms rather than feeling monotone throughout the experience. Next, a tactile device must provide a safe way to stream music sensations on the go, whilst not positioning the user into a place of danger. Interviews suggested a desire for music accessibility while driving, but noted the danger of looking at lyric screens. Lastly, the most successful devices would likely not act in isolation but would pair with visual apps to provide a more holistic experience.

Policy:

All interviews point to a centralized issue, not necessarily with solely a lack of music technology, but rather with a lack of government and policy support. Whether it be Sarahanne Dambrosio’s disencouraging music class, or Rosanna Ascenzo’s experience with faulty accessibility

technology, there are not enough policies which support the deaf experience when it comes to music. To create a more inclusive environment, government must impose stricter regulations regarding deaf accessible concerts, with screens for interpreters and qualified deaf communication professionals, as well as rules in schools regarding support for deaf children and students who want to be expose to music. Instead of turning them down, policies must provide these students with proper accessibility technology which will allow them to interpret arts in a different way than other students, while still allowing them to remain participative in class.

Conclusion

A head-worn, wireless vibration device is a highly viable solution for those seeking a personal, portable, and private way to "vibe" with music. However, the research indicates that it cannot be a total replacement for the lyrical and visual elements that many in the DHH community value most. As Ascenzo reminds us, "not all deaf people are the same... everybody has their own experiences". Therefore, such a device should be positioned as a tool for sensory enhancement rather than a singular solution for musical accessibility. Through the interviews, it is also clear that a lack of deaf accessible technology is not the only issue, but rather a lack in government support and policy which hides beneath the cover of the deaf community not understanding music.